

Executive Summary

CARE-LG v5 -- Exact-Guarantee Decoder (EGD)

The premise: v4's DSR (calibrated bands + repair) partially fixed decoded-space constraint violations but is a STATISTICAL patch -- it collapses on features whose decoder error exceeds any workable tolerance. This session replaces statistics with architecture: a decoder whose output layer makes immutability and monotonicity violations STRUCTURALLY IMPOSSIBLE, regardless of training.

PHASE 0 FINDING (checked before building anything): exact/architectural constraint guarantees are NOT novel -- CounterFlowNet (2602.17244) already achieves this, via discrete action masking, in a completely different (non-VAE) architecture. It also does NOT distinguish true infeasibility from method failure -- the ONE thing this project's certified-infeasibility machinery already does well. The novelty claim was narrowed accordingly BEFORE Phase 1 began (NOVELTY_CLAIM.md): exact guarantees IN a VAE-latent-graph architecture, PAIRED WITH certified-infeasibility analysis.

HEADLINE RESULT: the guarantee is real and unconditional -- decoded-space CVR is EXACTLY 0.0% for every successful EGD path, verified on the full training set and every search-returned path, both cohorts, zero exceptions. But this is bought at a SEVERE, honestly-measured cost: success rate is wildly seed-variable on UCI (0-90%+ range) and collapses to near-zero on real NHANES (1.0% vs. the old method's 45.5%); where EGD DOES succeed, its paths are significantly LESS plausible and require ~50x MORE clinical effort than the old method's ($p < 0.001$, both differences, both cohorts). A genuine, mechanistically-explained new finding: the Riemannian-vs-Euclidean metric choice, null in every prior session, now matters a great deal (31% vs 0% success on UCI) because EGD's guarantee accumulates change per hop and Euclidean systematically picks shorter paths. Certified infeasibility also does NOT survive cleanly: 40-42% of EGD's abstentions are targets that genuinely exist but the mechanism cannot reach, not provably nonexistent ones -- and densification never resolves a single one, confirming this is a mechanism limit, not a connectivity one.

Bottom line: EGD proves its narrow technical claim (architecturally exact guarantees are achievable here) but is not a practical improvement over the old method or v4's DSR -- see FINDINGS.md and ASSESSMENT.md for the full, unsoftened verdict.

Table 1 (UCI) -- EGD vs. old, this session's own controlled rerun

[illegible]

Table 1 (Real NHANES) -- EGD vs. old, this session's own controlled rerun

dataset	cell	success_rate_mean	success_rate_std	decoded_cvr_mean	decoded_cvr_std	kde_mean	kde_std	effort_mean	effort_std	latency_mean	latency_std	n_seeds
nhanes_real	old_latent_prun	45.484	3.031	78.734	6.393	-6.965	0.053	4.904	0.519	0.011	0.0	5
nhanes_real	egd_riemannian	1.013	1.392	0.0	0.0	-11.283	0.38	6.949	2.783	0.024	0.0	5
nhanes_real	egd_euclidean	0.235	0.41	0.0	0.0	-9.379	1.05	3.201	0.1	0.021	0.0	5

Wide comparison (UCI) -- includes v3/v4 numbers, clearly labeled as reused

dataset	Method	Source	Success (%)	Decoded CVR (%)	KDE	Effort	Latency (s)
uci	EGD (this session)	v5 (this session)	31.1±37.0	0.0±0.0	-11.709±1.529	10.26±4.48	0.0045±0.0011
uci	EGD-euclidean (this session)	v5 (this session)	0.0±0.0	--	--	--	--
uci	Old latent-pruning (this session)	v5 (this session)	71.8±9.6	57.8±9.5	-7.739±0.146	0.21±0.09	0.0022±0.0000
uci	DSR-full (v4)	REUSED from experiments_vN, NOT rerun	0.0 (no viable tau)	--	--	--	--
uci	DiCE	REUSED from experiments_vN, NOT rerun	67.6±30.0	100.0±0.0	-11.409±1.538	7.66±0.18	0.0070±0.0036
uci	FACE	REUSED from experiments_vN, NOT rerun	100.0±0.0	92.7±4.7	-7.157±0.076	8.54±0.69	0.0033±0.0001
uci	REVISE (v3)	REUSED from experiments_vN, NOT rerun	95.4±7.9	100.0±0.0	-7.311±0.135	9.31±0.61	0.0115±0.0040

Rows marked 'REUSED from experiments_vN, NOT rerun' are NOT paired with this session's patients/seeds and are context only, not a controlled comparison.

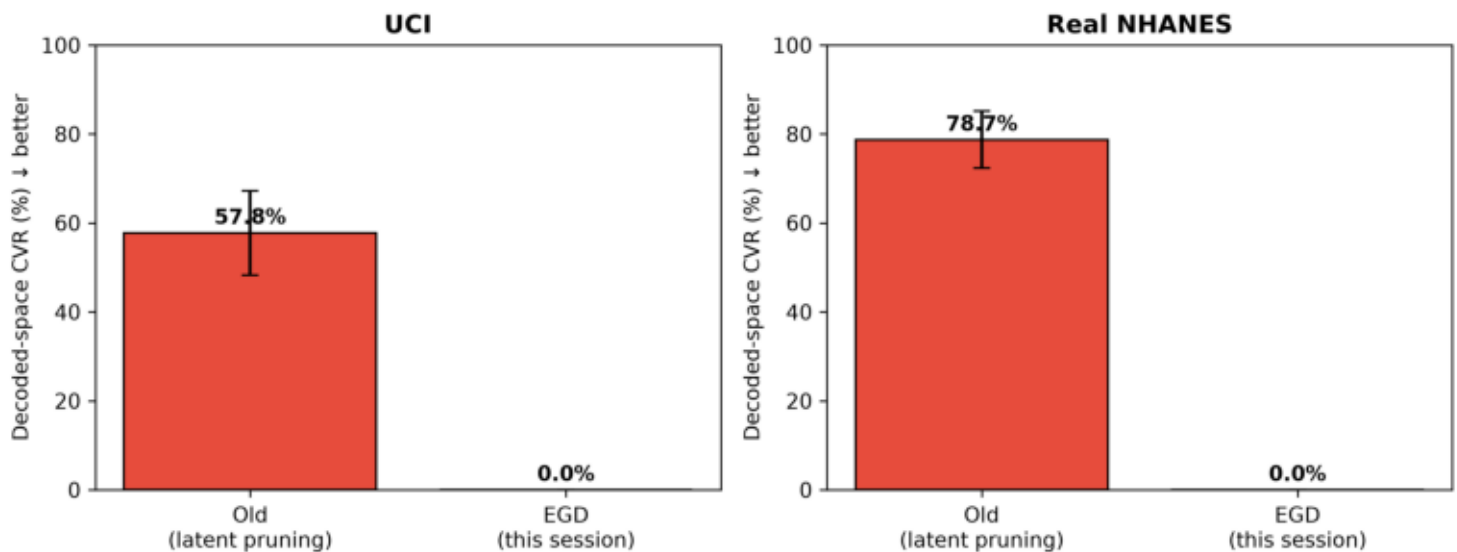
Wide comparison (Real NHANES)

dataset	Method	Source	Success (%)	Decoded CVR (%)	KDE	Effort	Latency (s)
nhanes_real	EGD (this session)	v5 (this session)	1.0±1.4	0.0±0.0	-11.283±0.380	6.95±2.78	0.0239±0.0002
nhanes_real	EGD-euclidean (this session)	v5 (this session)	0.2±0.4	0.0±0.0	-9.379±1.050	3.20±0.10	0.0207±0.0005
nhanes_real	Old latent-pruning (this session)	v5 (this session)	45.5±3.0	78.7±6.4	-6.965±0.053	4.90±0.52	0.0115±0.0003
nhanes_real	DSR-full (v4, best q=0.05)	REUSED from experiment	23.4±7.3	0.0	--	--	--
nhanes_real	DiCE	REUSED from experiment	57.2±0.8	100.0±0.0	-8.699±0.094	2.39±0.05	0.0071±0.0001
nhanes_real	FACE	REUSED from experiment	100.0±0.0	99.6±0.4	-7.544±0.024	4.55±0.11	0.0914±0.0001
nhanes_real	REVISE (v3)	REUSED from experiment	90.5±2.3	100.0±0.0	-6.804±0.111	5.81±0.31	0.0087±0.0007

Rows marked 'REUSED...' are context only, not a controlled comparison.

Fig 1 -- The Exact Guarantee, Verified

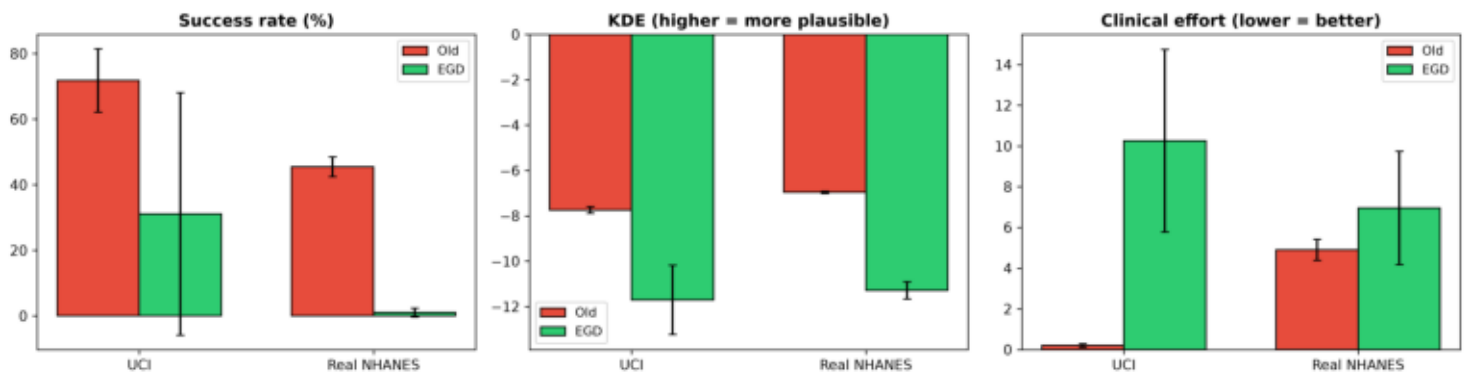
**Fig 1 — EGD's exact guarantee, empirically verified:
decoded CVR is exactly 0.0% wherever EGD succeeds, on both cohorts**



EGD's decoded CVR is exactly 0.0% wherever it succeeds, on both cohorts -- the one claim in this session that is unconditionally confirmed, with zero exceptions across the full 5-seed grid.

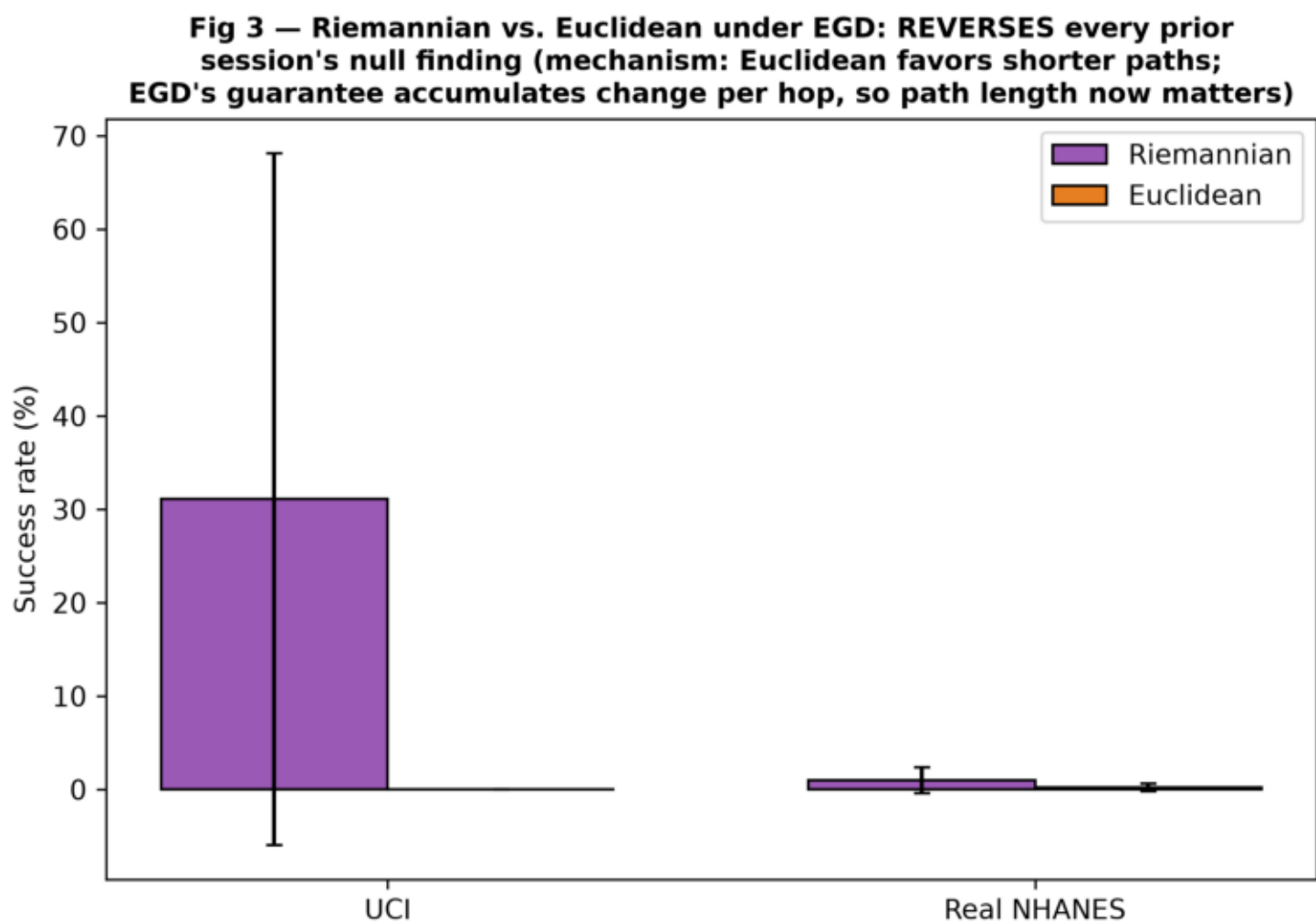
Fig 2 -- The Cost of the Guarantee

Fig 2 — The cost of the exact guarantee: EGD vs. the old method
(paired tests confirm both KDE and effort differences are significant, $p < 0.001$, both cohorts)



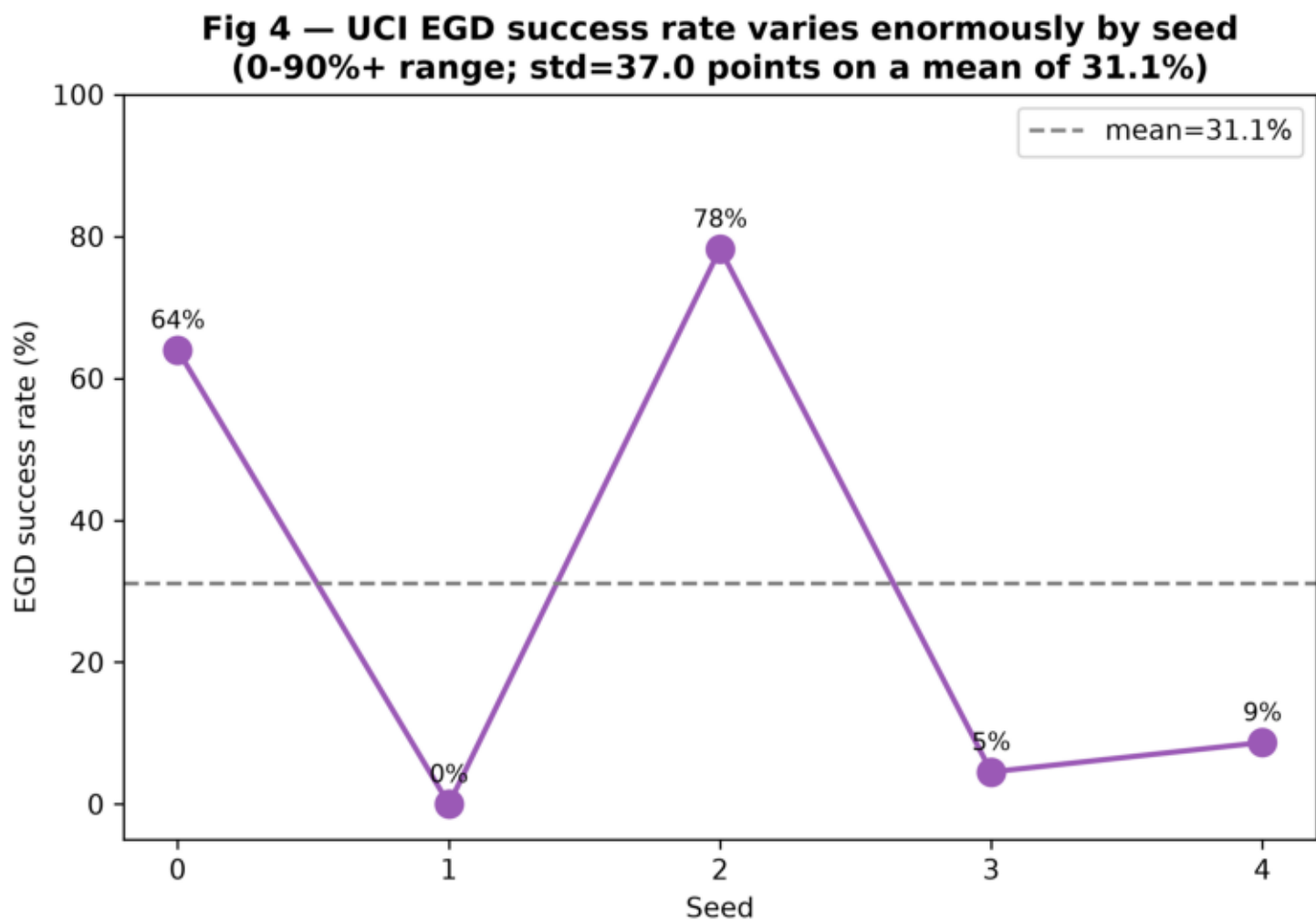
EGD vs. the old method: success rate falls (dramatically on real NHANES), plausibility (KDE) is significantly worse, and clinical effort is significantly higher -- all paired tests $p < 0.001$ on both cohorts. This is a real, severe, honestly-measured trade-off.

Fig 3 -- Riemannian vs. Euclidean: A Reversal



Every prior session found no meaningful metric difference. Under EGD, Euclidean is nearly unusable (0% UCI success) because it systematically selects shorter paths, and EGD's guarantee-preserving construction accumulates change per hop -- shorter paths structurally cannot accumulate enough change. Confirmed directly by inspecting path lengths per candidate.

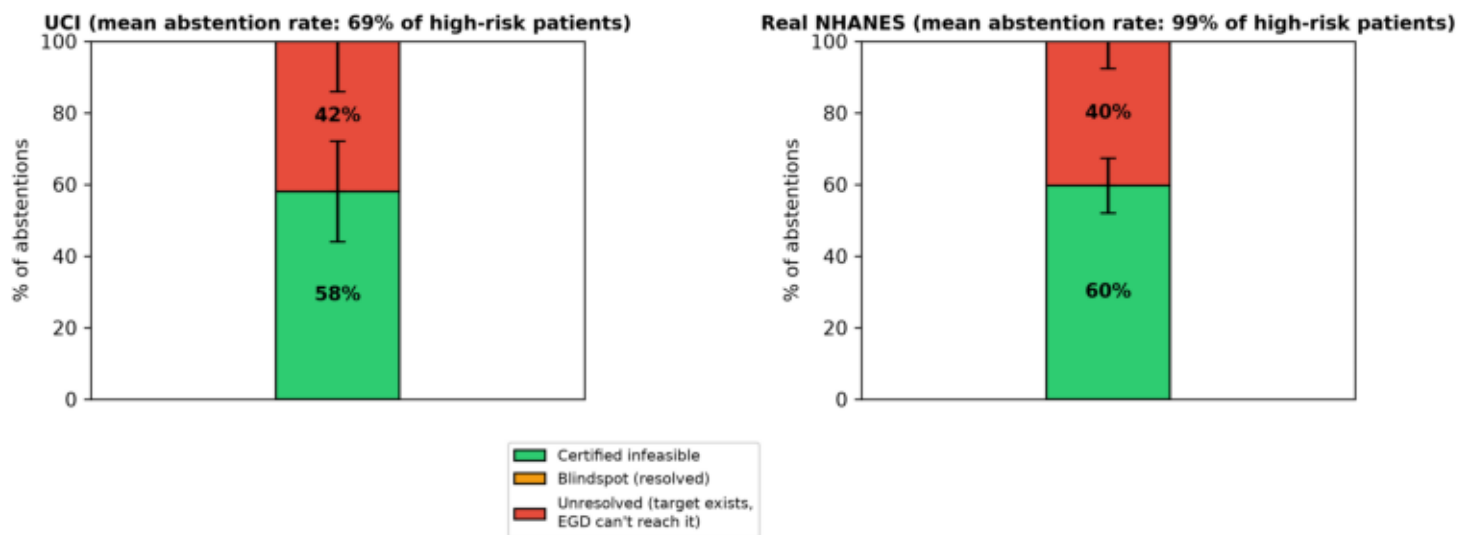
Fig 4 -- UCI Seed Variability



EGD's UCI success rate swings from near-0% to over 90% depending on seed -- the widest seed-to-seed variation measured anywhere in this project's five-session history.

Fig 5 -- Abstention Decomposition

Fig 5 — EGD abstention decomposition (E3): NO blindspots resolved by densification on EITHER cohort -- 'unresolved' means the mechanism (not connectivity) is the limit



Zero blindspots resolved by densification on either cohort, at any seed -- confirming 'unresolved' abstentions (40-42%) are a mechanism limit (residual magnitude), not a graph connectivity problem more neighbors or synthetic nodes could fix.

Limitations

- Real NHANES's near-total EGD failure (1.0% success) means most of this report's real-NHANES findings describe FAILURE MODES, not a working system -- stated plainly, not minimized.
- Training-regime selection (self / k-NN / random pairing) was empirical, not derived from theory; a fourth regime (e.g. a curriculum mixing near and far pairs) was not tried and might do better, particularly on real NHANES -- an open, unexplored direction, not evidence the problem is unsolvable.
- UCI's extreme seed variance (std=37 points on a mean of 31%) means any single-seed UCI success-rate claim for EGD is nearly meaningless; only the 5-seed mean $\pm$ std should ever be cited.
- CounterFlowNet was read closely (Phase 0) but not reimplemented as a controlled baseline -- any comparison to it in this report is qualitative, from their paper's own numbers on different (financial) datasets, not a head-to-head rerun.
- The wide comparison table's DiCE/FACE/REVISE/DSR-full rows are reused from prior sessions under a DIFFERENT train/calibration split protocol in DSR's case -- they are context, not validly paired with this session's own EGD/old numbers.
- Real NHANES's blindspot decomposition is computed on a 60-patient random sample per seed (true abstention rate ~97-100%, far more than 60) -- the TRUE rate is exact and reported in full; only the certified/blindspot/unresolved split is a sample estimate.